

## Use-Case Flow Specification

**Use Case Name:** UC-02: Parse EXPLAIN Plans & Recommend Indexes

**Primary Actor:** Database Administrator (DBA)

**Scope:** Database Query Performance Profiler

### Preconditions

1. The system has successfully ingested slow query logs from the target database instance.
2. The DBA is authenticated and has access to the observability dashboard.

### Postconditions

1. SQL execution plans are parsed and evaluated.
2. Index recommendations and scan warnings are rendered on the interface and stored in the database registry.

### Main Success Scenario

1. The Database Administrator selects a slow query entry from the observability dashboard.
2. The system fetches the raw EXPLAIN execution plan for the selected query from the host database.
3. The system parses the execution plan tree to evaluate scan types, join algorithms, and cost distribution.
4. The system identifies sequential table scans occurring on large data tables.
5. The system evaluates current table indexing schemas to locate missing or suboptimal index patterns.
6. The system generates recommended optimal composite index column definitions.
7. The system displays the visual execution tree alongside the recommended index definition to the DBA.

8. The use case ends successfully.

### **Alternate Flow**

- **4a. No Sequential Scans Detected / Query Already Optimized**
  - **4a1.** System determines execution latency is caused by system resource contention (e.g., CPU/Memory lock waits) rather than missing indexes.
  - **4a2.** System flags the query as "Schema Optimized" and displays host system metric bottlenecks instead of index suggestions.
  - **4a3.** The use case ends successfully.